

Solution Requirements

Date	13 July 2026
Team ID	N/A
Project Name	FinRelief AI – AI Powered Debt Relief & Financial Recovery Platform
Maximum Marks	4 Marks

Define Functional and Non-Functional Requirements:

The following tables define exactly what the FinRelief AI solution will do (Functional Requirements) and how well it will perform (Non-Functional Requirements).

Step 1: Functional Requirements (FR)

S. No	Requirement Category	Requirement Description	Priority
1	Authentication	Users securely log in using email and password, verified via JWT-based token authentication.	High
2	Authorization levels	Two roles: Borrower (manages own loans and negotiations) and Admin (oversees platform data and AI activity logs).	Medium
3	External interfaces	Integrates with the Google Gemini API for AI-generated negotiation letters and settlement recommendations.	High
4	Transactions processing	Loan data submissions, financial health calculations, and AI requests are processed synchronously through FastAPI endpoints.	High
5	Reporting	The dashboard generates financial health reports showing EMI ratio, DTI ratio, monthly surplus, and settlement predictions.	Medium
6	Business rules, etc.	Settlement recommendations are bounded by configurable minimum/maximum percentages based on overdue duration and income.	Medium
7	Compliance to laws or regulations	Borrower financial data is handled in line with standard data privacy practices; no data is shared with third parties beyond the Gemini API call.	High
8	Other	Rule-based fallback recommendations are provided if the Gemini API is temporarily unavailable.	Low

Step 2: Non-Functional Requirements (NFR)

S. No	NFR Category	Requirement Description	Target Metric / Acceptance Criteria
1	Performance & Speed	Dashboard and negotiation pages should load and respond quickly for a smooth borrower experience.	Page loads in < 2 seconds
2	Scalability	The system should handle growth in registered borrowers and loan records without major redesign.	Supports up to 1,000 concurrent users
3	Security & Data Privacy	Borrower financial and personal data is protected through encryption and secure authentication.	JWT-based auth with hashed passwords; HTTPS in production
4	Reliability & Availability	The platform should remain available with minimal downtime, and gracefully fall back if the AI service fails.	99% uptime; rule-based fallback on AI failure
5	Usability & Accessibility	The interface should be simple and clear enough for financially-stressed users to navigate without confusion.	Clean, guided UI with plain-language financial terms
6	Other	The codebase should be maintainable and portable across environments.	Modular FastAPI services; environment-based configuration